

OSSP-5

Question:

Implement a producer-consumer communication system using anonymous pipes where the parent process generates data and the child process consumes it. Measure the communication efficiency.

Develop a program that executes the equivalent of the shell command: `ls -l | grep ".c"` using `fork()`, `pipe()`, `dup2()`, and `exec()` system calls

Solution:

```
ubuntu@ubuntu:~$ nano producer_consumer.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <string.h>
#include <sys/wait.h>
#include <time.h>

int main() {
    int fd[2];
    char buffer[100];

    pipe(fd);

    clock_t start = clock();

    pid_t pid = fork();

    if(pid > 0) {
        close(fd[0]);

        char data[] = "Hello from Producer (Parent Process)";
        write(fd[1], data, strlen(data)+1);

        close(fd[1]);
        wait(NULL);
    }
    else {
        close(fd[1]);

        read(fd[0], buffer, sizeof(buffer));

        printf("Consumer received: %s\n", buffer);

        close(fd[0]);
        exit(0);
    }

    clock_t end = clock();

    double time_taken =
        ((double)(end-start))/CLOCKS_PER_SEC;

    printf("Communication Time: %f seconds\n",
        time_taken);

    return 0;
}

ubuntu@ubuntu:~$ gcc producer_consumer.c -o producer_consumer

ubuntu@ubuntu:~$ ./producer_consumer
Consumer received: Hello from Producer (Parent Process)
Communication Time: 0.000296 seconds

ubuntu@ubuntu:~$ ls -l | grep ".c"
drwxr-xr-x 2 ubuntu ubuntu 40 Sep  9 06:55 Documents
drwxr-xr-x 2 ubuntu ubuntu 40 Sep  9 06:55 Music
drwxr-xr-x 2 ubuntu ubuntu 40 Sep  9 06:55 Pictures
drwxr-xr-x 2 ubuntu ubuntu 40 Sep  9 06:55 Public
-rwxrwxr-x 1 ubuntu ubuntu 70936 Sep  9 07:07 producer_consumer
-rw-rw-r-- 1 ubuntu ubuntu 802 Sep  9 07:06 producer_consumer.c
ubuntu@ubuntu:~$ nano pipe_ls_grep.c
```

```

#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    int fd[2];
    pipe(fd);
    pid_t pid1 = fork();
    if(pid1 == 0) {
        dup2(fd[1], STDOUT_FILENO);
        close(fd[0]);
        close(fd[1]);
        execlp("ls", "ls", "-l", NULL);
        perror("exec ls failed");
        exit(1);
    }
    pid_t pid2 = fork();
    if(pid2 == 0) {
        dup2(fd[0], STDIN_FILENO);
        close(fd[1]);
        close(fd[0]);

```

```

        execlp("ls", "ls", "-l", NULL);
        perror("exec ls failed");
        exit(1);
    }
    pid_t pid2 = fork();
    if(pid2 == 0) {
        dup2(fd[0], STDIN_FILENO);
        close(fd[1]);
        close(fd[0]);
        execlp("grep", "grep", ".c", NULL);
        perror("exec grep failed");
        exit(1);
    }
    close(fd[0]);
    close(fd[1]);
    wait(NULL);
    wait(NULL);
    return 0;
}

```

```

ubuntu@ubuntu:~$ gcc pipe_ls_grep.c -o pipe_ls_grep
ubuntu@ubuntu:~$ ./pipe_ls_grep
drwxr-xr-x 2 ubuntu ubuntu 40 Sep  9 06:55 Documents
drwxr-xr-x 2 ubuntu ubuntu 40 Sep  9 06:55 Music
drwxr-xr-x 2 ubuntu ubuntu 40 Sep  9 06:55 Pictures
drwxr-xr-x 2 ubuntu ubuntu 40 Sep  9 06:55 Public
-rw-rw-r-- 1 ubuntu ubuntu 674 Sep  9 07:11 pipe_ls_grep.c
-rwxrwxr-x 1 ubuntu ubuntu 70936 Sep  9 07:07 producer_consumer
-rw-rw-r-- 1 ubuntu ubuntu 802 Sep  9 07:06 producer_consumer.c

```